Shinde Rajvardhan Sunil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.85
Sanjay Ghodawat Junior College, Atigre	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2024	86.67%
Sitabai Patwardhan Highschool, Kurundwad	Class X (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2022	95.40%

EXPERIENCE:

IITH Racing Team: Steering Subsystem Member

- Optimized steering rack location and tie-rod hard points through suspension kinematic analysis to minimize bump steer across suspension travel.
- Refined steering linkage geometry and scrub radius to improve steering response, cornering stability, and tyre wear characteristics.

Drone Building Project:

(Prof. Vishnu Rajasekharan Unni, IITH)

- Assembled a quadcopter by integrating the frame, motors, ESCs, Flight Controller, receiver.
- Performed soldering, wiring and hardware integration of electronic and mechanical components.

Particle Image Velocimetry (PIV): (Ongoing)

(Prof. S. K. Karthick, IITH)

- Designing a Particle Image Velocimetry (PIV) setup for quantitative flow-field measurements.
- Finalised the key components required for the imaging, illumination, and particle-seeding systems.
- Will integrate and calibrate the setup, followed by PIV experiments and flow-field analysis for velocity measurement.

PROJECTS:

Steering Geometry Solver using Matlab (Ongoing)**:** Link

- Developing a MATLAB-based steering geometry solver using the Pacejka '96 tyre model to predict tyre forces, slip angles.
- Developing a steady-state cornering model incorporating aerodynamic downforce and drag to simulate realistic race car behavior.
- Developing the solver to simulate very close to real-world vehicle behaviour by moving beyond the bicycle model and ideal 100% Ackermann methods.

Dynamic Analysis of a Multi-Cylinder IC Engine (Matlab)**:** Link

- Developed a baseline dynamic model to analyze shaking forces and torque demand of multi-cylinder inline engines.
- Enhanced the model by incorporating approximate combustion gas force and piston-cylinder friction for a production diesel engine.

- Calculated bearing reactions, piston side thrust, shaking forces, and engine torque using Newton-Euler dynamics.

Planar Mechanism Synthesis & Motion Validation using Solidworks: Link

- Synthesized a planar mechanism using the three-position synthesis method to achieve the prescribed motion of a partially circular object.
- Designed the mechanism using revolute and prismatic joints while satisfying the specified quick-return ratio for the return stroke.
- Developed and validated the mechanism in SolidWorks to verify the required positions and timing.

Formula 1 Aerodynamics: Engineering Analysis Link

- Authored a comprehensive technical report analyzing the evolution of Formula 1 aerodynamics from the origins of aerodynamic wings to the modern ground-effect era.
- Performed 2D steady-state and transient CFD simulations of flow around a circular cylinder using ANSYS Fluent to demonstrate vortex formation and wake development for explaining fundamental aerodynamic phenomena.
- Analyzed F1's clever innovations including blown diffusers, the evolution of DRS from the F-duct, bargeboards, ground effect, flexi-wings, and porpoising.
- Studied airflow management strategies for brake cooling, engine cooling, tyre wake control, and CFD-wind tunnel correlation to understand modern Formula 1 aerodynamic design philosophies.

TECHNICAL SKILLS:

- Programming Language: C, Python, Matlab
- CAD: Solidworks, Solid Edge
- Simulation: Solidworks Simulation (FEA), Ansys CFD

ACHIEVEMENTS:

- Secured A+ in Engineering Drawing, Fabrication Laboratory.
- Secured A in Engineering Mechanics, Fluid Mechanics, Digital Fabrication.

EXTRACURRICULAR ACTIVITIES & INTERESTS:

- Interested in motorsports engineering, with a focus on technical knowledge.
- Passionate about Formula 1, WRC, and WEC, with a strong interest in their innovations and technical regulations.
- Enjoy cycling, badminton, and driving.